AI-use journal

Annex B | Week 1 | 12–13 November 2026

I used AI to explore options, write search code and develop the experiment. I made the final design choices based on the project scope and the pilot results. The instructions below are short summaries, not full prompts.

Date	Tool / version	Task and instruction	Output	My evaluation and correction	What I kept
12 Nov	ChatGPT / GPT-6	Give me three ways to build a fibroid assistant and compare their research potential.	Standard RAG, evidence-aware selective RAG and graph-based QA.	I chose the second direction. A simple chatbot is not enough for my research question, and a graph needs too much medical annotation.	Scope and three-way comparison.
12 Nov	ChatGPT / GPT-6	Help me turn my medical and technical keywords into database queries and Python search code.	Search code and query versions for PubMed, Europe PMC and OpenAlex.	I kept medical evidence and technical papers separate. I compared the actual search counts and retained the raw records.	Versioned search equations and pilot data.
12 Nov	ChatGPT / GPT-6	Compare the results of the different queries and suggest which ones to keep.	Title relevance categories and proposed query choices.	I noticed a mismatch between the purpose of a query and some returned titles. MED-v3, for Tunisia, even returned a study in Qatar. I kept this query only as a local supplement.	MED-v2/v5 as main searches; MED-v3 as supplement.
12 Nov	ChatGPT / GPT-6	Find related methods for citation support, uncertainty and medical RAG.	A list of candidate references and research directions.	I treated the list as leads, not verified evidence. A matching title is not enough: the paper must answer my question and the identifier must match the original source.	Candidate references and a narrower research question.

Example: why the query result is not enough

MED-v3 was intended to find evidence from Tunisia. One returned title was *Uterine fibroids in women of childbearing age: a cross-sectional study at the primary health care corporation in Qatar.* This is a concrete example of why a database hit cannot be treated as a local study. I need to check the actual study location and population, not just the keywords or author affiliation. MED-v3 returned 17 records; none was directly relevant by the preliminary title coding.

Technical decisions and evaluation

Date	Tool / version	Task and instruction	Output	My evaluation and correction	What I kept
12–13 Nov	ChatGPT / GPT-6	Help me organize the system variants for a fair comparison.	M0–M3 and possible combinations of retrieval, ranking and verification.	I chose to keep M1 as plain hybrid RAG rather than an enhanced version that already uses evidence-quality ranking. Otherwise the effect of ranking in M2 would be unclear.	M1 = BM25 + embeddings + RRF; M2 adds ranking; M3 adds checking and a gate.
12–13 Nov	ChatGPT / GPT-6	Explore how to measure the contribution of each component and possible failure cases.	Candidate ablations, metrics and eight FibroidQA scenarios.	I want to test the gate separately from claim checking. I also want to compare systems at the same coverage, because a system that refuses everything can look safe.	Ablation plan, risk–coverage evaluation and draft questions.
Next step	Claude / version to record	Independent check of the protocol, references and experimental logic against the source documents.	Review checklist prepared; no completed Claude output recorded.	Check any findings against the original papers and actual search data, not against another AI answer.	Record only findings and corrections after the check.

1. What was wrong or incomplete, and how did I detect it?

Some returned titles did not match the purpose of my search. The Qatar paper in the Tunisia search was a good example. I realized that a title can also look relevant while the actual population or outcome is different. I kept the title labels as preliminary and will check abstracts, study locations and original references before making inclusion decisions.

2. What decision did I make differently?

I considered an M1 that included both hybrid retrieval and evidence-quality ranking. I prefer to keep M1 as normal hybrid RAG and move the evidence ranking to M2. M3 then adds claim checking and abstention. This way, if the results improve, I can test which part made the difference. I also want to compare a gate-only version with a checking-only version.

3. What would I do differently without AI, and what did I learn?

Without AI, I would spend more time writing search queries and exploring existing methods. I learned that combining techniques is not enough to make a research contribution. I need a clear hypothesis, controlled baselines and a way to measure failures. For example, I want to measure unsupported claims at the same coverage rather than give credit to a model just because it refuses more often.